

40-Minute Defense Speech — Mouad Louhichi

How to use this script. Each block is the spoken text for a slide (or a group of slides), with the target slide number and a running time. The whole script is timed to land

in or around 38–40 minutes, leaving a margin for the jury's interruptions and the Q&A.

Words in [brackets] are stage directions or emphasis cues — do not read them aloud.

Pauses are marked as (pause).

Total: ~5,800 words $\approx$ 38 min at a measured ~150 words/min. If you run long, the two marked "CUT IF TIGHT" segments are the safest places to drop.

Introduction (Slides 3-6) — ~4 min

Slide 3 — Title & hello (0:00-0:30)

Good morning. Thank you, President and Professors, for the time you are giving me to present

and discuss my thesis. My name is Mouad Louhichi. My thesis is titled Cooperative Game

Theory for Explainable Artificial Intelligence in Recommendation Systems: A Shapley Framework

for Actionable Insight, supervised by Professor Mohamed Lazaar. (pause) In one sentence,

it advances this idea: Shapley-value attribution is not just a post-hoc explanation — it is a

single formal mechanism that can explain black-box clustering, stay coherent under hierarchical scale, and finally operate as an in-training signal inside a recommender.

Slide 4 — Motivation: three questions (0:30-1:30)

Three questions frame the whole work. First, ubiquity. Opaque systems mediate what billions of people see, buy and watch every day — not only on streaming platforms, but in

e-commerce, news feeds and search. Second, the black box. Even strong recommenders and

clustering pipelines are hard to interrogate. We observe their outputs without understanding

why a particular item was chosen. Third, trust. Transparency should be built into the modelling logic, not bolted on afterwards. (pause) The core tension of the thesis is that as

models gain expressive power, they lose the transparency needed for trustworthy deployment.

Throughout this work I hold accuracy and interpretability as objectives to be reconciled,

not traded against one another.

Slide 5 — Actionable insight (1:30-2:30)

Let me define the frame. I call an explanation actionable when it identifies at least one

modifiable factor whose change is associated with a specifiable change in model output —

and when that factor is expressible in the semantic vocabulary of the task domain: a physicochemical variable for wine, a pollution indicator for air quality, a preference signal

for recommendation. The crucial constraint is that it must not be an opaque latent code.

This distinction separates explanations that let a designer intervene and act from those

that merely describe what happened. By this definition, a credible explanation is judged by

whether it supports a downstream decision, not by whether it looks plausible.

Slide 6 — Research context (2:30-3:30)

The context is a progression from simple to hypergraph recommenders. Matrix factorisation

showed that latent factors could capture preferences — but those factors were immediately

uninterpretable. Graph and graph-neural-network models improved ranking by exploiting

connectivity, yet replaced an opaque latent code with an opaque message-passing mechanism.

Finally, hypergraph models added higher-order user-item-context relations, but typically assume

that every message contributes uniformly. (pause) Each step raised expressiveness

while

lowering transparency. The deficit matters for trust, debugging and regulation — the EU AI

Act, the OECD principles and the GDPR all put a premium on meaningful explanation. That is why

a principled attribution mechanism is needed, and why I will argue it should be part of the

modelling logic rather than an afterthought.

Context & Problematic (Slides 7-12) — ~5 min

Slide 7 (section) + Slide 8 — Paradigms (3:30-4:30)

Quick orientation across the paradigms we build on. Collaborative filtering recommends on the

principle that similar users will value similar items. Content-based filtering recommends items

sharing attributes with a profile, while hybrid methods combine both. Matrix factorisation

factorises the interaction matrix into latent factors — compact and effective, but immediately

opaque. Graph-based methods, including LightGCN and hypergraph extensions, propagate

information over an interaction structure and capture multi-hop or higher-order relations. Each

strengthens the modelling, but each complicates interpretation in a specific way: matrix

factorisation made latent dimensions opaque; graph models kept importance implicit; hypergraph

models added higher-order relations but assumed uniform message importance. That uniformity

assumption is one of the things I challenge.

Slide 9 — Limitations of classical models (4:30-5:15)

Four classical limitations. Data sparsity and scalability are the most cited: the user-item

matrix is overwhelmingly empty. Cold-start is the structural consequence — a new user or

item has no history to learn from. Popularity bias and the lack of diversity create a

filter-bubble loop. And the most fundamental limitation, and the one this thesis targets, is

the absence of interpretability. For clustering specifically it is even harder: methods privilege a local explanation or a global one but not both; they struggle to scale; and their explanations rarely remain coherent across resolutions.

Slide 10 — Problem statement (5:15-6:00)

Three structuring problems. First, lack of explainability. Second, difficulty of scaling — local explanations do not transfer naturally to hierarchical structures or large

datasets. Third, weak integration into learning — most explanations are post-hoc and do not

shape model dynamics, nor the accuracy-diversity-context trade-off. The thesis gap follows

directly: the literature still lacks a single cooperative-attribution framework that explains

clustering faithfully, stays coherent under hierarchy, and then operates as an in-training

signal in recommendation. My claim is that Shapley-value attribution can be that framework.

Slide 11 — Research questions (6:00-6:45)

Five questions form the spine. RQ1: how can Shapley values explain black-box clustering

faithfully at instance and cluster level? RQ2: how can that extend to large-scale,

hierarchical clustering without losing tractability or consistency? RQ3: can cooperative

attribution move beyond post-hoc analysis into the learning dynamics of graph recommenders?

RQ4: can a recommender jointly optimise ranking, context and diversity when importance is

estimated by a cooperative-game utility? And RQ5, the thesis-level question: what emerges

when clustering explanation and recommendation learning are read as two stages of one shared

perspective?

Slide 12 — Three contributions (6:45-7:30)

Three contributions, one thread. C1 establishes Shapley-based explanation for

black-box

clustering through a PCA-K-Means-LightGBM-TreeSHAP pipeline, validated on wine. C2 scales

it to hierarchy and large data via multi-level clustering with cross-level SHAP aggregation,

validated on Beijing air quality. C3 is the strongest claim: DyHuCoG replaces post-hoc attribution with an in-training signal inside a hypergraph recommender, validated on MovieLens-1M and Amazon-Book. Each answers a distinct research question, but they are designed

to be read together as a cumulative argument — cooperative game theory as a shared

attribution perspective for explanation, optimisation and intervention.

Experimental Protocol (Slides 13-17) — ~2.5 min

Slide 14 — Datasets (7:30-8:20)

Two clustering datasets and two recommendation datasets. For clustering I deliberately picked

datasets with semantically interpretable features, because the whole point of the method is

to return attribution to the original variables. Wine Quality is a small, dense, chemically

correlated dataset of almost five thousand samples with eleven features. Beijing Multi-Site Air

Quality is a large, noisy, temporally and meteorologically variable dataset of over three

hundred and eighty thousand hourly records. For recommendation I used benchmark-standard

datasets with established baselines: MovieLens-1M, with roughly a million interactions and a

density of 0.0447, and Amazon-Book, which is far sparser at 0.0006. The sparsity contrast is

deliberate — it stress-tests whether a Shapley-guided model helps most precisely when

interaction data are weak.

Slide 15 — Splitting (8:20-8:50)

Splitting is designed against leakage. For clustering I use five-fold cross-validation to

test

the stability of the surrogate and the attribution. For recommendation I use a user-level,

temporal holdout — seventy percent train, ten percent validation, twenty percent test — with

leave-one-out evaluation, where the latest test positive per user is the target. Because

MovieLens contains explicit ratings, I convert ratings greater than three into positive implicit feedback, and draw negatives from a popularity-aware distribution to produce harder contrasts. Results are reported across five seeds, with early stopping on validation

NDCG.

Slide 16 — Baselines & metrics (8:50-9:30)

Baselines span classical, neural, graph and hypergraph methods, so I isolate the contribution

of cooperative attribution rather than a favourable model choice. For recommendation I compare

against MF, NCF, LightGCN, RecDCL, HCCF and HPCF, treating HPCF as the strongest reference. For

clustering interpretability I compare against a LIME-based surrogate. Ranking is measured with

Precision, Recall and NDCG at twenty as the principal measure. On diversity I measure

catalogue coverage and intra-list diversity, defined as the average pairwise

dissimilarity inside a ranked list. I want to emphasise that ILD is not decorative — it is deliberately built into the DyHuCoG coalition utility. Finally, every headline comparison is

backed by statistical validation.

Slide 17 — Hardware (9:30-9:55)

The hardware matters mainly because it explains some runtime figures I will quote. All

clustering, preprocessing and data loading run on an Intel Core i9 with twenty-four cores.

DyHuCoG training and inference run on an NVIDIA RTX 4090 with twenty-four gigabytes, alongside

forty-eight gigabytes of RAM. The stack is Python 3.8, scikit-learn, LightGBM, SHAP

and PyTorch

2.0.1. Everything stays within ordinary academic compute.

Contribution I (Slides 18-31) — ~5 min

Slide 19 (objectives) + 20 (RQ1→objectives) (9:55-10:40)

Why start with clustering? Here the model creates its own structure, so cluster meaning must be

inferred after the fact — which makes clustering the hardest and most natural test bed for

attribution. The gap is that explainable clustering is fragmented: methods favour a local or a

global explanation, rarely both; they often fail to scale; and they rarely preserve coherence

across clusters. Because Shapley explanation is well established in supervised problems, its

near absence from unsupervised clustering is striking. RQ1 decomposes into three objectives:

O1, a pipeline yielding cluster-level explanation while preserving feature-level attribution;

O2, to preserve the semantics of the original feature space rather than a latent space; and O3,

to justify why Shapley is the right concept rather than an ad-hoc surrogate such as LIME.

Slide 21 — Cooperative-game framing (10:40-11:20)

We frame clustering as a cooperative game where features are the players. The value function

measures how well the data cluster when we use only the features in a given coalition: the value

of coalition S is the Silhouette of K-Means run on the feature subset, with fixed k . I choose

Silhouette because it is bounded, normalised and semantically intuitive. A feature then receives

a high Shapley value when its presence consistently improves separation across all coalitions.

The problem is that evaluating Silhouette for every subset is combinatorial and infeasible —

which is exactly why we need a bridge between the unsupervised partition and a tractable attribution method.

Slide 22 — The surrogate bridge (11:20-12:05)

The bridge is the heart of the method. TreeSHAP explains tree models; it cannot explain K-Means

centroids. And explaining the PCA representation would move attribution away from the

interpretable variables we care about. So we convert an unsupervised partition into a supervised

task: once K-Means produces cluster labels, we train a LightGBM multiclass surrogate to

predict them from the original features, then apply TreeSHAP to that surrogate. This keeps

attribution in the original semantic feature space, which is what makes the explanation

actionable. The key validity condition is surrogate fidelity — we require a macro-F1 of around 0.82 as the floor.

Slide 23 — Pipeline (12:05-12:40)

The pipeline has five stages. Standardisation, then PCA as a computational and visual diagnostic — deliberately not the explanatory space. Then K-Means-plus-plus with multi-criteria k selection. Then the LightGBM surrogate. Then TreeSHAP attribution in the

original feature space, aggregated into global importance, cluster-specific profiles, and local

force plots. Complexity is dominated by PCA and repeated K-Means; TreeSHAP scales with tree

count and depth rather than exponentially in features — which is what makes the approach

tractable.

Slide 24 — k selection (12:40-13:20)

This is an honest and important point. We evaluate k from two to ten, and deliberately select

k equal to three even though it is not geometrically optimal. With k equal to two we get a

stronger Silhouette of 0.214 and a Davies-Bouldin of 1.775; with k equal to three the

Silhouette

drops to 0.144. On raw geometry, two is the better partition. We choose three nonetheless,

because three clusters support three distinct, chemically meaningful oenological narratives — far

more actionable than a dichotomous split. I select the partition a domain expert would find

useful, not the one that maximises a separation index. And to avoid a common confusion: the 0.63

Silhouette belongs to the Beijing dataset in Contribution II, not this one.

Slide 25 — Global importance (13:20-13:55)

The global SHAP ranking is dominated by density, followed by pH, fixed acidity, sulfur-dioxide-related variables and alcohol. These are precisely the variables an oenologist

would point to as governing structure, preservation and sensory balance. The important point is

that this is not an arbitrary classifier artefact — because the surrogate was trained to reproduce the partition from the original variables, and because TreeSHAP attributes in that

original space, the recovered hierarchy corresponds to a chemically interpretable structure.

This is the strongest evidence that the pipeline is faithful: it recovers domain knowledge

without being told to.

Slide 26 — Cluster profiles (13:55-14:30)

The cluster-specific profiles show the solution is not only globally interpretable but internally differentiated. Cluster zero is driven by density and sulfur-dioxide variables; cluster one by acidity and pH; cluster two by a different balance across acidity, alcohol and

related chemicals. Crucially, the same small set of variables recurs across all three clusters,

but with different relative weights within each. This is the actionable insight: a cluster is

not just a label, it is a distinct, domain-meaningful combination of drivers.

Slide 27 (Answers) + 28 (Key findings) (14:30-15:15)

To answer RQ1 directly: yes. Shapley values explain a black-box partition faithfully,

provided the surrogate is high-fidelity, and coherently at cluster level. Each objective is met.

O1 is met because the pipeline yields both a global and a per-cluster reading. O2 is met because

attribution is returned to the original chemical variables. O3 is met because Shapley satisfies

efficiency, symmetry, the null-player property and additivity, whereas LIME has no equivalent

guarantee. The honest caveat is that efficiency holds with respect to the surrogate output, not

directly to the Silhouette-based game.

Slide 29 — Limitations (15:15-15:40)

Let me be clear about the scope. First, the explanation depends on the fidelity of the LightGBM

surrogate; it is not a direct mechanism of the K-Means geometry. Second, the approach is confined

to tabular data. Third, and most importantly for what follows, it is single-level: it cannot

yet address hierarchical coherence, meaning it cannot explain how feature importance

reconfigures between a partition and its sub-partitions. These limits define the point of

departure for Contribution II.

Slide 30 — Takeaways (15:40-16:00)

The takeaway is that Shapley attribution works as a single, principled lens for explaining an

unsupervised partition, and that keeping attribution in the original feature space is what makes

it actionable. But real data are rarely single-level — broad regimes contain nested sub-groups.

That flat limitation directly motivates Contribution II: can the explanation logic be scaled to

multi-level, large-scale clustering without losing coherence?

Contribution II (Slides 32-45) — ~5 min

Slide 33 (obj) + 34 (RQ2) (16:00-16:45)

C2 asks whether the C1 logic survives scale and hierarchy. Large real-world data contain

structure at more than one granularity: broad regimes at the top, nested sub-groups within them.

A variable can be globally important yet locally uninformative, so a flat explanation is true but

incomplete. The chapter adds three things. First, a genuine multi-level workflow. Second, a

formal cross-level consistency argument, Proposition 6.1. Third, validation on a structurally

different large-scale dataset, Beijing air quality.

Slide 35 — Multi-level architecture (16:45-17:30)

The architecture proceeds recursively: coarse clustering on the full dataset, then subdivide each

cluster where appropriate, producing a nested structure. For each level we train a level-specific

surrogate and compute SHAP in the same original feature space. The cross-level aggregation is

deliberately not a naive average — it respects cluster size and nesting structure. A

parent-level attribution is an expectation over the explanatory structure of its descendants. One

methodological point: the hierarchy here is a pragmatic analytical device. I am not claiming

the data have a true ontological hierarchy; the nested structure is a computational and

interpretive tool.

Slide 36 — Proposition 6.1 (17:30-18:15)

Proposition 6.1 is the mathematical heart. Let Φ_l at level l in cluster c for feature j be the

expected absolute SHAP importance over examples in that cluster, and let $w_{c'}$ be the

relative size of a child within its parent. For a strict nested hierarchy on a consistent feature

space, the parent's expected absolute importance equals the sum over its children of the child's

relative size times the child's expected absolute importance, plus a residual epsilon that comes from surrogate mismatch. The derivation uses the law of total expectation. Crucially, it does not claim explanations are identical across levels; it says differences can be interpreted rather than dismissed as inconsistency — which makes a hierarchical explanation self-consistent and auditable.

Slide 37 — Evaluation on Beijing (18:15-18:50)

On Beijing, the multi-criteria evaluation converges much more strongly than on wine. The full dataset clusters cleanly into k equal to three, with a Silhouette of about 0.63 and a Davies-Bouldin of about 0.55. Recall this is where the 0.63 figure comes from — it belongs to Beijing, not to the wine partition. I also tested sensitivity: the conclusions are robust to modest variation in k , in projection dimension, and in surrogate depth; only low-ranked variables shift.

Slide 38 — Global importance (Beijing) (18:50-19:30)

The dominant features are temperature, dew point and pressure, followed by carbon monoxide, nitrogen dioxide, PM10 and PM2.5. It is striking that it is not simply the pollutant concentrations that dominate. Meteorological variables play a structurally central role, because temperature, dew point and pressure condition dispersion, trapping and photochemical behaviour. This is exactly the kind of insight flat, descriptive summaries often fail to make explicit.

Slide 39 — Regimes (19:30-20:10)

The force plots reveal three representative regimes. Regime A is a warm, photochemical regime, where ozone, temperature and dew point are prominent — characteristic of summer photochemical

smog. Regime B is a wintertime smog regime, dominated by carbon monoxide, sulfur dioxide and particulate matter, with low wind speed suppressing dispersion. Regime C is a comparatively clean-air regime associated with favourable meteorology. The interpretative value is not merely showing that these regimes exist — it is showing which combinations of variables define each one.

Slide 40 — Multi-level insight (20:10-20:50)

This is the conceptual payoff. At the coarse level, temperature and dew point dominate, because they differentiate broad atmospheric regimes. Within individual clusters, the discriminative variables shift to carbon monoxide, sulfur dioxide, PM10, wind speed, pressure or ozone. This change is not a contradiction — it is exactly what a multi-level explanation should reveal. The parent-level story is regime selection; the cluster-level story is variation within a regime. Proposition 6.1 lets us interpret these shifts as meaningful structure rather than noise.

Slide 41 — Cross-dataset generality (20:50-21:25)

The same logic works on both a small, dense, chemically correlated dataset and a large, noisy, environmentally variable one, which supports the generality of the approach — it is not tied to a domain-specific peculiarity of wine chemistry. Compared against the SHAP-based clustering literature, our Beijing partition achieves a Silhouette of about 0.63 versus around 0.37 for Gramegna and Giudici on credit-risk. On LIME we observe weaker structural coherence and less stable local narratives for hierarchical reasoning. I want to be careful: these are comparative observations rather than a comprehensive benchmark.

Slide 42 (Answers) + 43 (Key findings) (21:25-22:05)

To answer RQ2 directly: yes, with bounds. Shapley-based clustering explanation can scale to hierarchical, large-scale settings without losing interpretive coherence, provided the hierarchy is modelled explicitly and the approximation is transparent. O1, O2 and O3 are all met. The honest bound is that the model is still an explanation of a pre-computed partition — it does not yet influence learning itself. That is the bridge to Contribution III.

Slide 44 — Limitations (22:05-22:30)

The clustering remains static even though the Beijing data are temporal. The surrogate-based SHAP plus representative-instance reporting compresses observation-level variation. The approach is confined to tabular data. And, most importantly, the model is still an explanation of a pre-computed partition: it does not yet influence learning itself.

Slide 45 — Takeaways (22:30-22:50)

Shapley attribution stays coherent across granularity when the hierarchy is made explicit. But the attribution is still post-hoc: it explains a partition that was already computed, and it never influences how the model learns. That is the limit that motivates Contribution III — can cooperative attribution move beyond explanation and enter the learning dynamics of a recommender?

Contribution III — DyHuCoG (Slides 46-64) — ~7.5 min

Slide 47 (obj) + 48 (RQ3/RQ4) (22:50-23:45)

C3 is the flagship contribution and the conceptual shift of the whole thesis. The gap is that graph and hypergraph recommenders treat message importance as either uniform or attention-weighted, without a principled account of marginal contribution. Diversity is typically a secondary objective or a re-ranking heuristic. And interpretability is added

after

prediction rather than integrated into the objective. Our objectives are threefold: formulate

recommendation as a cooperative game; embed preference-aware Monte Carlo Shapley into hypergraph

message passing; and improve ranking, coverage and diversity jointly. The strongest claim is not

that we explain a recommender — it is that attribution becomes an in-training signal.

Slide 49 — Cooperative-game framing (23:45-24:30)

We model recommendation as a cooperative game whose players are users, items and contexts. The

hypergraph has vertices from this same set, and hyperedges encode user-item-context

interactions, with dynamic edge weights derived from Shapley estimates. A coalition represents

the entities participating in a recommendation episode, and the coalition value measures the

quality of the recommendation outcome. This parallels the clustering formulation, but with a

recommendation-oriented value function. That parallelism is deliberate, and is one of the

thesis-level contributions.

Slide 50 — Coalition utility (24:30-25:15)

The coalition utility combines ranking quality, diversity and context: a weighted sum of NDCG at

twenty, a diversity term and a context score, with weights summing to one. The key point is that

this is the same trade-off the recommender must satisfy and also the trade-off from which

attribution is computed — so the explanatory game and the predictive objective are aligned by

design. Alpha, beta, gamma were grid-searched to 0.60, 0.25, 0.15. One honest boundary: coalition

evaluation is scoped to the interaction episode, a few dozen players, not the full catalogue.

Slide 51 — Monte Carlo Shapley (25:15-26:00)

Exact Shapley is combinatorial, so we use a Monte Carlo estimator that averages, over M sampled coalitions, the difference between the value including a player and the value excluding it; the preference-aware variant applies the same estimator to the preference-weighted utility. It is unbiased, with variance decaying as σ^2/M . We chose M equal to fifty: mean squared error about one point four times ten to the minus five, and roughly ninety-nine percent accuracy. Estimates are refreshed every ten batches and smoothed with an exponential moving average, so attribution is adaptive without making training hypersensitive to a single estimate.

Slide 52 — Architecture (26:00-26:50)

The architecture is the decisive move. The base propagation is standard hypergraph message passing; the Shapley-weighted version weights each message by a normalised Shapley coefficient. I clip and exponentially smooth the estimates before normalising, which stabilises the sparse regime. Layer fusion combines embeddings across layers with learned coefficients. On top of that, an interaction-level attention gate interpolates between the Shapley-weighted score and the standard inner-product score, acting as a stabiliser; a context-aware score adds a context term. The essential point is that the model is told not only who is connected to whom, but how much each coalition is worth, and that worth directly governs how information propagates.

Slide 53 — Loss (26:50-27:35)

The composite loss combines four terms: a Bayesian Personalised Ranking loss; an intra-list diversity regulariser; a context alignment term; and weight decay. Negatives are drawn from a popularity-aware distribution with periodic hard-negative refresh, and optimisation uses Adam. The conceptual point I want to drive home is alignment: the model is trained to

optimise the same accuracy-diversity-context balance that later determines cooperative attribution. So the explanation is not a separate diagnostic — it is a direct read-out of the objective the model is already optimising. This is what gives the explanation its structural faithfulness.

Slide 54 — Headline results (27:35-28:30)

This is the headline result. On MovieLens-1M, the strongest baseline HPCF achieves NDCG at twenty of 0.2528 and recall of 0.2098; DyHuCoG improves these to 0.2775 and 0.2362 — plus nine point seven seven percent in NDCG and plus twelve point five eight percent in recall. At the same time, coverage rises from 0.342 to 0.397 and intra-list diversity from 0.461 to 0.516. On Amazon-Book, which is much sparser, the relative gains are larger: plus thirteen point three percent NDCG and plus sixteen point one six percent recall, with coverage up by nearly thirty percent. The sparser the data, the larger the gain — precisely the pattern you would expect if Shapley-driven weighting is most valuable when signal is weak.

Slide 55 — Relative gains (28:30-29:05)

This summarises the relative gains over the strongest baseline, and the key message is that DyHuCoG improves ranking accuracy, coverage and diversity simultaneously — it does not sacrifice one for the others. Together these results are evidence that the accuracy-diversity trade-off, often treated as structurally fixed, is in fact negotiable if attribution is handled as a first-class part of the learning objective.

Slide 56 — Diversity (29:05-29:40)

Both levels of diversity improve. On MovieLens-1M, catalogue coverage rises by sixteen percent and intra-list diversity by eleven point nine percent. On Amazon-Book, coverage rises by nearly

thirty percent. The practical consequence is a reduced filter-bubble effect and greater discovery

opportunity. And I want to stress: both diversity metrics improve while NDCG and recall also

improve — so we are not trading accuracy for diversity.

Slide 57 — Ablation (29:40-30:30)

This ablation isolates each component, and every component contributes. Removing the Shapley

value drops NDCG by four point six percent. Removing the hypergraph structure costs six point

eight percent. Removing the attention gate costs three point five percent. Removing context

causes the largest single loss at eight point two percent. Removing diversity costs five point

eight percent. The four to six percent drop when Shapley is removed supports the argument that

marginal-contribution estimation is load-bearing rather than decorative.

Slide 58 — Computational cost (30:30-31:10)

The attribution cost is proportionate, and I report it transparently because it bears on deployability. DyHuCoG takes about two thousand seconds to train versus roughly eleven hundred

for HPCF — about one point seven eight times. Inference is one point eight four milliseconds per

query on MovieLens, comfortably within real-time requirements. Memory is modestly higher. On

Shapley convergence, M equal to fifty gives ninety-nine percent accuracy, while M equal to one

hundred gives ninety-nine point five with diminishing returns — so fifty is the right operating

point.

Slide 59 — Statistical validation (31:10-31:50)

I validated that the improvements are statistically significant and substantively meaningful with

paired t-tests on per-user NDCG at twenty across six thousand and forty users. Against HPCF, the

t-statistic is 46.38, Cohen's d is 1.33, and the p-value is on the order of ten to the

minus two
hundred and seventy. The Wilcoxon signed-rank test is also significant. I should be scrupulous
about scope: these tabulated paired tests apply only to MovieLens-1M; the Amazon-Book results
remain descriptive.

Slide 60 — Cold-start & interpretability (31:50-32:35)

DyHuCoG also improves the regimes where recommenders are at their most brittle. For users with
five or fewer interactions, NDCG at twenty is about 0.061, roughly ten percent over HPCF; for
cold-start items about 0.057. On interpretability, the model produces a SHAP waterfall that
decomposes a recommendation into the same ranking, diversity, context and preference components
used during training — so the explanation is structurally faithful rather than an external
approximation. And Shapley measures marginal utility rather than raw frequency, so weak but
informative interactions retain influence, mitigating popularity bias.

Slide 61 (Answers) + 62 (Key findings) (32:35-33:20)

To answer RQ3 and RQ4 directly: yes. RQ3 is answered because attribution is no longer a
post-hoc diagnostic but an in-training signal that directly shapes message passing. RQ4 is
answered because ranking, coverage and diversity improve together rather than trading against one
another. The strongest claim is that the explanation is structurally faithful, because it reads out the same components the model already optimises. Within the stated edges, the claim
stands.

Slide 63 — Limitations (33:20-33:50)

The limitations must be stated honestly. There is a measurable computational overhead, roughly
one point seven eight times the training time of HPCF. The method depends on the availability of

meaningful context. The Monte Carlo Shapley estimator could be improved by variance reduction. The ablation is component-wise, so it does not test factorial interactions. And the baselines were finalised in early 2026, so I claim superiority only against the tested baselines. Within those bounds, the claim stands.

Slide 64 — Takeaways (33:50-34:15)

The takeaway is the conceptual shift of the thesis. Attribution is a first-class part of the learning objective, not a post-hoc diagnostic, and the explanation is a direct read-out of the objective the model already optimises. That is what makes it structurally faithful. And because Shapley values rest on an axiomatic, normative basis, this perspective is well aligned with the transparency and accountability requirements of emerging AI regulation. (pause) This is the central claim worth defending: explanation should be part of the modelling logic itself.

Conclusion & Perspectives (Slides 65-72) — ~3.5 min

Slide 66 — Synthesis (34:15-34:55)

Three contributions, one thread. C1 makes hidden structure intelligible. C2 keeps that explanation coherent under scale and hierarchy. C3 carries the same attribution logic inside the learning dynamics of a recommender. The common thread is that cooperative attribution is a single mechanism used for explanation, for interpretation and for optimisation — not three unrelated tools.

Slide 67 — Published papers (34:55-35:25)

The thesis synthesises three peer-reviewed publications: Paper I in *Procedia Computer Science* 2023, Paper II in *IJACSA* 2025, and Paper III, *DyHuCoG*, in *IJIES* 2026. What the thesis

adds on top

of the papers is the multi-level formalisation, the cross-chapter comparisons, and the explicit

mapping to the five research questions. In other words, the thesis is a cumulative argument,

not merely a bound collection of three papers.

Slide 68 — Limitations (35:25-35:55)

Let me state the limitations honestly. Computationally, exact Shapley is intractable.

Methodologically, the clustering chapters depend on surrogate fidelity, and the recommendation

results depend on stable approximate contributions and meaningful context.

Empirically, the work

is confined to tabular data and offline evaluation — no multimodal, sequential or online

deployment, and no dedicated human-subject study of actionability. And on the claim itself: I am

not claiming one fully unified framework that eliminates all tension between accuracy and

interpretability; I am claiming a shared perspective, a common attribution language, coherent and

productive across the three tasks.

Slide 69 — Perspectives (35:55-36:25)

Future work turns these limitations into a concrete agenda. First, scalable cooperative attribution, with lower-variance Shapley estimators and adaptive refresh. Second, online and

streaming recommendation with evolving graphs and delayed feedback. Third, richer human-centred

evaluation — do these explanations measurably improve analyst judgement, user trust,

intervention quality or perceived fairness? And fourth, broader trustworthy-AI and fairness

evaluation, connecting the work to the EU AI Act, the OECD principles and the GDPR.

Slide 70 — Conclusion (36:25-37:00)

To close: the thesis answer is that cooperative game theory can function as a shared methodological perspective for actionable explanation across clustering and recommendation. The

key outcomes are fourfold: Shapley attribution provides a common formal language for allocating importance to features, interactions and contexts; it yields faithful clustering explanation, hierarchical explanatory coherence, and contribution-aware recommendation learning; and it moves explanation from commentary to method — from post-hoc description to in-training guidance. I believe this is the central claim worth defending.

Slides 71-72 — References (37:00-37:20) (CUT IF TIGHT)

These are the core references that ground the argument — the original Shapley value, the SHAP framework and its tree extension, the LIME and XAI taxonomy work, the key recommender and hypergraph baselines, and the regulatory anchor of the EU AI Act. I will be happy to go deeper on any of them during the questions.

Slide 73 — Q&A (37:20)

Thank you very much for your attention. I am now happy to take your questions and comments — I will do my best to answer them directly, and I am happy to go deeper into any of the methodology, the mathematics, or the experimental details.

Timing summary

Section	Slides	Target time	Running total
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Introduction	3-6	4 min	4 min
Context & Problematic	7-12	5 min	9 min
Experimental Protocol	13-17	2.5 min	11.5 min
Contribution I	18-31	5 min	16.5 min
Contribution II	32-45	5 min	21.5 min
Contribution III	46-64	7.5 min	29 min
Conclusion & Perspectives	65-72	3.5 min	32.5 min
Q&A opening	73	—	—

Target: 32-33 minutes of talking + interruptions $\approx$ 38-40 min. The two "CUT IF TIGHT" segments let you reclaim $\sim$ 1.5 min.